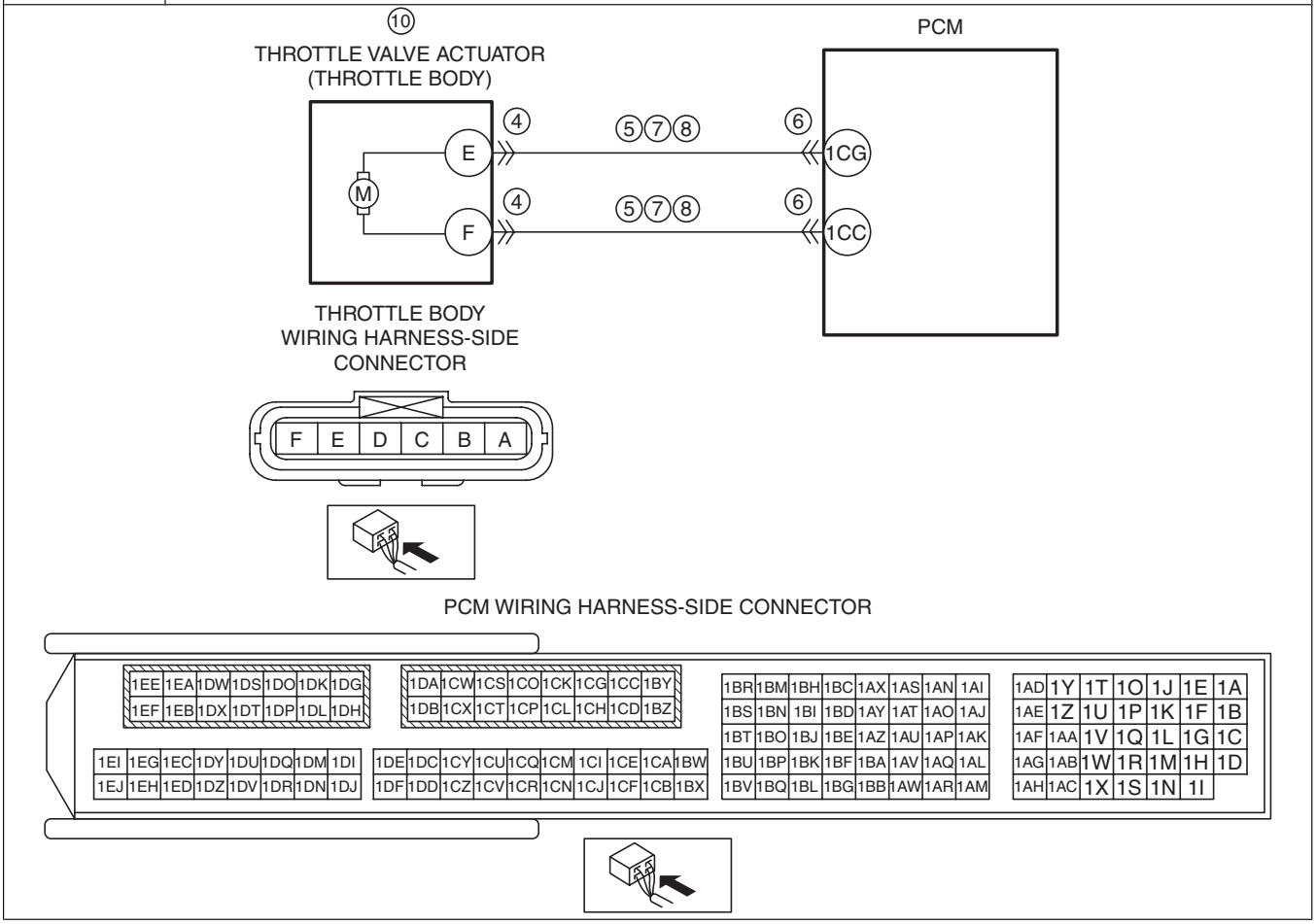
## DTC P2109:00 [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

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| DTC P2109:00 | TP sensor minimum stop range/performance problem |
|---|---|
| DETECTION CONDITION | • If the fully-close throttle position is **5.1 ° or lower**, or **15.9 ° or more** (even though fully-closed throttle position learning is finished), the PCM determines there is a malfunction.<br>**Diagnostic support note**<br>• This is a continuous monitor (CCM).<br>• The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle.<br>• FREEZE FRAME DATA/Snapshot data is available.<br>• DTC is stored in the PCM memory. |
| FAIL-SAFE FUNCTION | • Restricts the upper limit of the engine speed.<br>• Stops the drive-by-wire control (throttle valve is open at **approx. 8 °** by return spring force). |
| POSSIBLE CAUSE | • Throttle body connector or terminals malfunction<br>• Short to ground in wiring harness between the following terminals:<br>— Throttle body terminal E—PCM terminal 1CG<br>— Throttle body terminal F—PCM terminal 1CC<br>• PCM connector or terminals malfunction<br>• Short to power supply in wiring harness between the following terminals:<br>— Throttle body terminal E—PCM terminal 1CG<br>— Throttle body terminal F—PCM terminal 1CC<br>• Open circuit in wiring harness between the following terminals:<br>— Throttle body terminal E—PCM terminal 1CG<br>— Throttle body terminal F—PCM terminal 1CC<br>• Throttle valve malfunction<br>• Throttle valve actuator malfunction<br>• PCM malfunction |



**Diagnostic Procedure**

| STEP | INSPECTION | | ACTION |
|---|---|---|---|
| 1 | **RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION**<br><br>**Note**<br>• Recording can be facilitated using the screen capture function of the PC.<br><br>• Record the FREEZE FRAME DATA/snapshot data on the repair order. | — | Go to the next step. |
| 2 | **VERIFY RELATED SERVICE INFORMATION AVAILABILITY**<br>• Verify related Service Information availability.<br>• Is any related Service Information available? | Yes | Perform repair or diagnosis according to the available Service Information.<br>• If the vehicle is not repaired, go to the next step. |
| | | No | Go to the next step. |
| 3 | **VERIFY RELATED PENDING CODE AND/OR DTC**<br>• Switch the ignition off, then ON (engine off).<br>• Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)<br>• Are any other PENDING CODEs and/or DTCs present? | Yes | Go to the applicable PENDING CODE or DTC inspection. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) |
| | | No | Go to the next step. |
| 4 | **INSPECT THROTTLE BODY CONNECTOR CONDITION**<br>• Switch the ignition off.<br>• Disconnect the throttle body connector.<br>• Inspect for poor connection (such as damaged/ pulled-out pins, corrosion).<br>• Is there any malfunction? | Yes | Repair or replace the connector and/or terminals, then go to Step 11. |
| | | No | Go to the next step. |
| 5 | **INSPECT THROTTLE VALVE ACTUATOR CIRCUIT FOR SHORT TO GROUND**<br>• Verify that the throttle body connector is disconnected.<br>• Inspect for continuity between the following terminals (wiring harness-side) and body ground:<br>— Throttle body terminal E<br>— Throttle body terminal F<br>• Is there continuity? | Yes | If the short to ground circuit could be detected in the wiring harness:<br>• Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals:<br>— Throttle body terminal E—PCM terminal 1CG<br>— Throttle body terminal F—PCM terminal 1CC<br>**If there is a common connector:**<br>— Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground.<br>— Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>— Repair or replace the wiring harness which has a short to ground.<br>If the short to ground circuit could not be detected in the wiring harness:<br>• Replace the PCM (short to ground in the PCM internal circuit).<br>(See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)<br>Go to Step 11. |
| | | No | Go to the next step. |
| 6 | **INSPECT PCM CONNECTOR CONDITION**<br>• Disconnect the PCM connector.<br>• Inspect for poor connection (such as damaged/ pulled-out pins, corrosion).<br>• Is there any malfunction? | Yes | Repair or replace the connector and/or terminals, then go to Step 11. |
| | | No | Go to the next step. |

| STEP | INSPECTION | | ACTION |
|---|---|---|---|
| 7 | **INSPECT THROTTLE VALVE ACTUATOR CIRCUIT FOR SHORT TO POWER SUPPLY**<br>• Verify that the throttle body and PCM connectors are disconnected.<br>• Switch the ignition ON (engine off).<br><br>**Note**<br>• Another DTC may be stored by the PCM detecting an open circuit.<br>• Measure the voltage at the following terminals (wiring harness-side):<br>— Throttle body terminal E<br>— Throttle body terminal F<br>• Is the voltage **0 V**? | Yes | Go to the next step. |
| | | No | Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals:<br>• Throttle body terminal E—PCM terminal 1CG<br>• Throttle body terminal F—PCM terminal 1CC<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has a short to power supply.<br>Go to Step 11. |
| 8 | **INSPECT THROTTLE VALVE ACTUATOR CIRCUIT FOR OPEN CIRCUIT**<br>• Verify that the throttle body and PCM connectors are disconnected.<br>• Switch the ignition off.<br>• Inspect for continuity between the following terminals (wiring harness-side):<br>— Throttle body terminal E—PCM terminal 1CG<br>— Throttle body terminal F—PCM terminal 1CC<br>• Is there continuity? | Yes | Go to the next step. |
| | | No | Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals:<br>• Throttle body terminal E—PCM terminal 1CG<br>• Throttle body terminal F—PCM terminal 1CC<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has an open circuit.<br>Go to Step 11. |
| 9 | **INSPECT THROTTLE VALVE**<br>• Perform the Electronic Control Throttle Operation Inspection.<br>(See ENGINE CONTROL SYSTEM OPERATION INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)<br>• Is there any malfunction? | Yes | Repair or replace the malfunctioning part according to the inspection results, then go to Step 11. |
| | | No | Go to the next step. |
| 10 | **INSPECT THROTTLE VALVE ACTUATOR**<br>• Inspect the throttle valve actuator.<br>(See THROTTLE BODY INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)<br>• Is there any malfunction? | Yes | Repair or replace the malfunctioning part according to the inspection results, then go to the next step. |
| | | No | Go to the next step. |
| 11 | **VERIFY DTC TROUBLESHOOTING COMPLETED**<br>• Always reconnect all disconnected connectors.<br>• Clear the DTC from the PCM memory using the M-MDS.<br>(See AFTER REPAIR PROCEDURE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)<br>• Perform the KOEO or KOER self test.<br>(See KOEO/KOER SELF TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)<br>• Is the same DTC present? | Yes | Repeat the inspection from Step 1.<br>• If the malfunction recurs, replace the PCM.<br>(See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)<br>Go to the next step. |
| | | No | Go to the next step. |
| 12 | **VERIFY AFTER REPAIR PROCEDURE**<br>• Perform the “AFTER REPAIR PROCEDURE”.<br>(See AFTER REPAIR PROCEDURE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)<br>• Are any DTCs present? | Yes | Go to the applicable DTC inspection.<br>(See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) |
| | | No | DTC troubleshooting completed. |